



Massachusetts Institute of Technology

Anyonic Neural Network Wavefunctions

A.P. Fadon, P.M. Tam, **L. Shackleton**, arXiv:2610.xxxxx

L. Shackleton, arXiv:2610.xxxxx

T. Wang, **L. Shackleton**, M. Metlitski, in progress

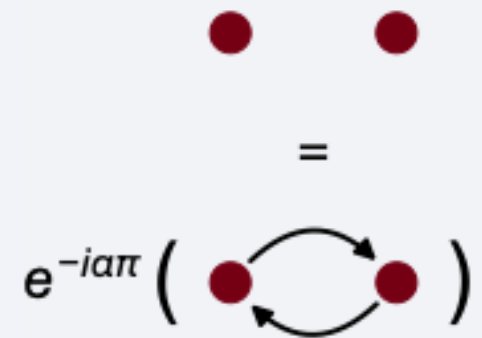
Generalized statistics in anyonic particles



Bosons



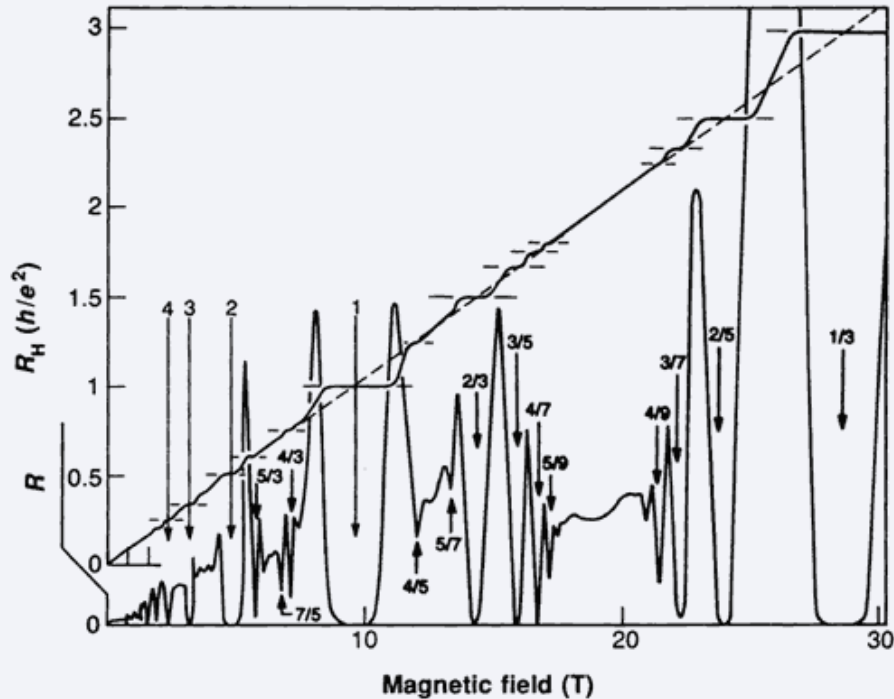
Fermions



Anyons

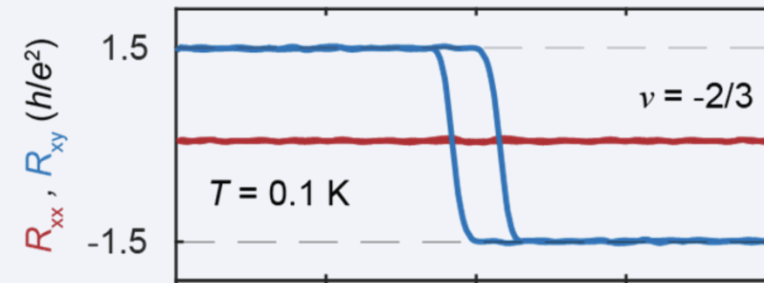
Anyons can be realized in physical systems

Fractional Quantum Hall

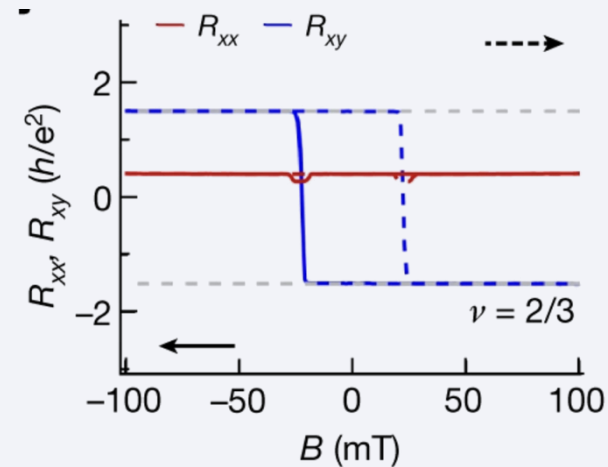


Tsui, Störmer, and Gossard, '80
Figure: Eisenstein and Störmer, '90

Fractional Quantum Anomalous Hall



Twisted MoTe2
Park et al, '23



Multilayer graphene
Lu et al, '24

Zero field, anyons acquire **dispersion**

Many proposals for anyons at finite density

Shi and Senthil, '25

Zhang, '26

Nosov, Han, and Khalaf, '26

Shi and Senthil, '26

Pichler, Kuhlenkamp, Knap, and Vishwanath, '26

Shi, Zhang, and Senthil, '25

Lotrič and Simon, '26

Wang, '26

Senthil, '26

Han, Vishwanath, and Khalaf, '26

Mehta, Nakajima, and Goldman, '26

...

Many superconductors

Holon metal

Vortex glass

Orthogonal metal

Charge density wave

Pair density wave

Anyon crystal

...

Outcome will likely depend on **microscopic details**

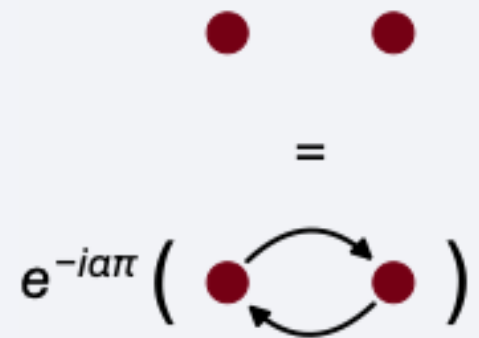
Effects of statistics at finite density



Bose condensation



Pauli exclusion,
Fermi gas



??????

”Free” anyons are strongly interacting

$$H = -\frac{1}{2m} \sum_i \nabla_i^2 \quad \psi(\mathbf{r}_1, \dots, \mathbf{r}_i, \mathbf{r}_j, \dots, \mathbf{r}_N) = e^{-i\alpha\pi} \psi(\mathbf{r}_1, \dots, \mathbf{r}_j, \mathbf{r}_i, \dots, \mathbf{r}_N)$$

Equivalent to (via gauge transformation)

$$H = -\frac{1}{2m} \sum_i [\nabla_i - i\mathbf{a}_i]^2 \quad \psi(\mathbf{r}_1, \dots, \mathbf{r}_i, \mathbf{r}_j, \dots, \mathbf{r}_N) = \psi(\mathbf{r}_1, \dots, \mathbf{r}_j, \mathbf{r}_i, \dots, \mathbf{r}_N)$$

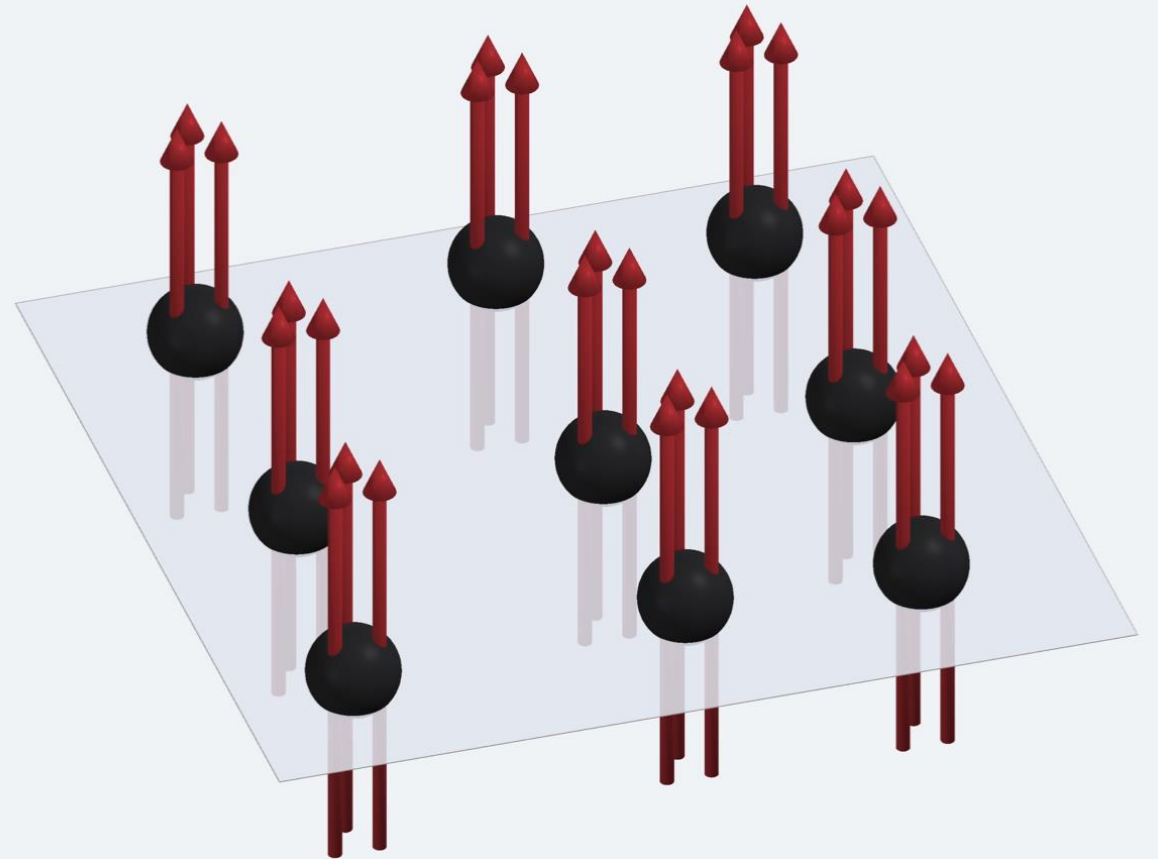
$$\mathbf{a}_i \equiv \alpha \sum_{j \neq i} \frac{\hat{\mathbf{z}} \times (\mathbf{r}_i - \mathbf{r}_j)}{|\mathbf{r}_i - \mathbf{r}_j|^2} \quad \nabla \times \mathbf{a}_i = 2\pi\alpha \hat{\mathbf{z}} \sum_{j \neq i} \delta^2(\mathbf{r}_i - \mathbf{r}_j)$$

Flux attachment and superfluidity

$$\nabla \times \mathbf{a}_i = 2\pi\alpha \hat{z} \sum_{j \neq i} \delta^2(\mathbf{r}_i - \mathbf{r}_j)$$

Smear out magnetic field to be uniform

Near the fermionic limit,
 $\alpha = 1(1 - 1/n)$



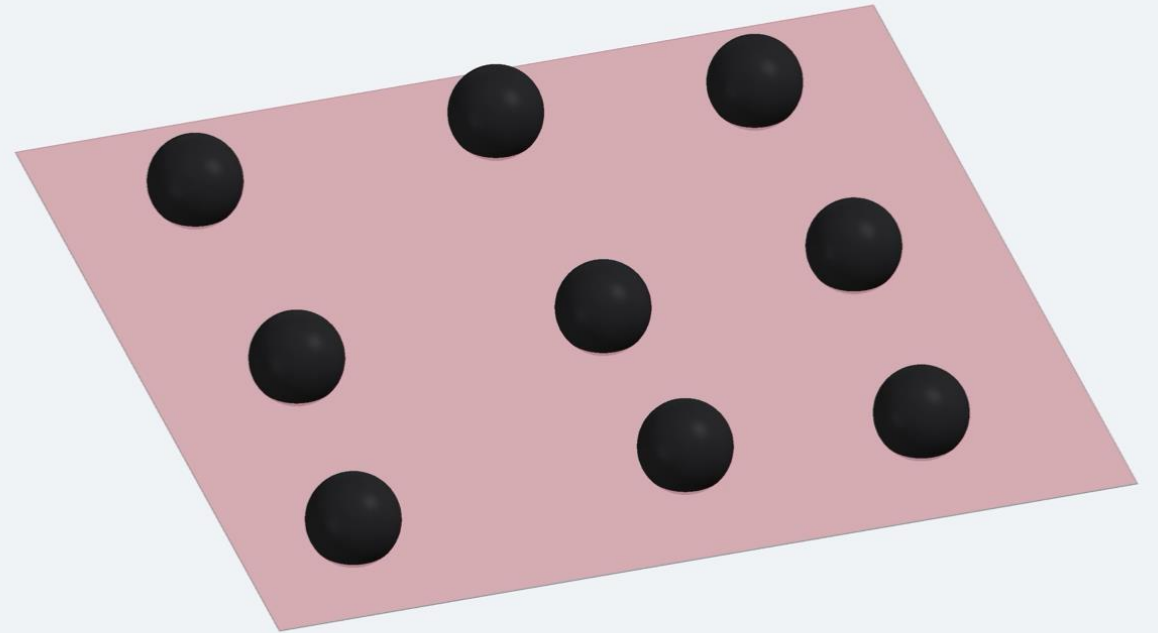
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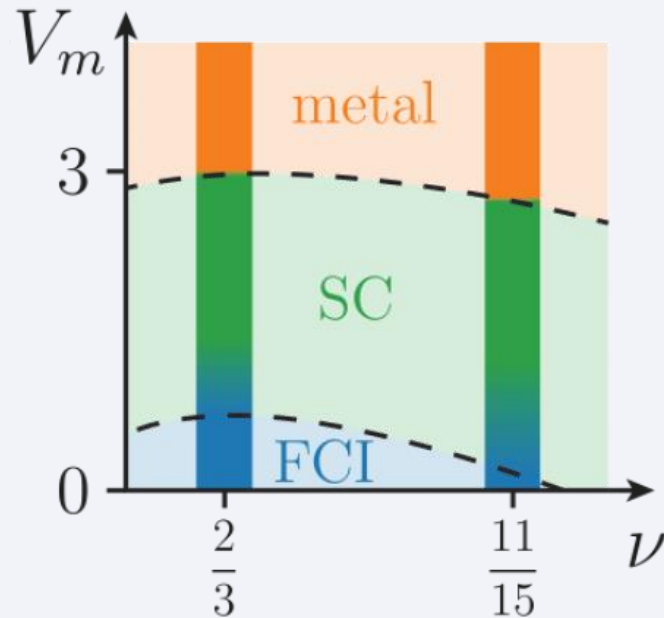
Near the fermionic limit,
 $\alpha = 1(1 - 1/n)$

Landau level filling: $\nu = \frac{2\pi\rho}{b} = n$



Anyons are difficult to simulate numerically

Anyons in FQH



Wang and Zaletel, '25

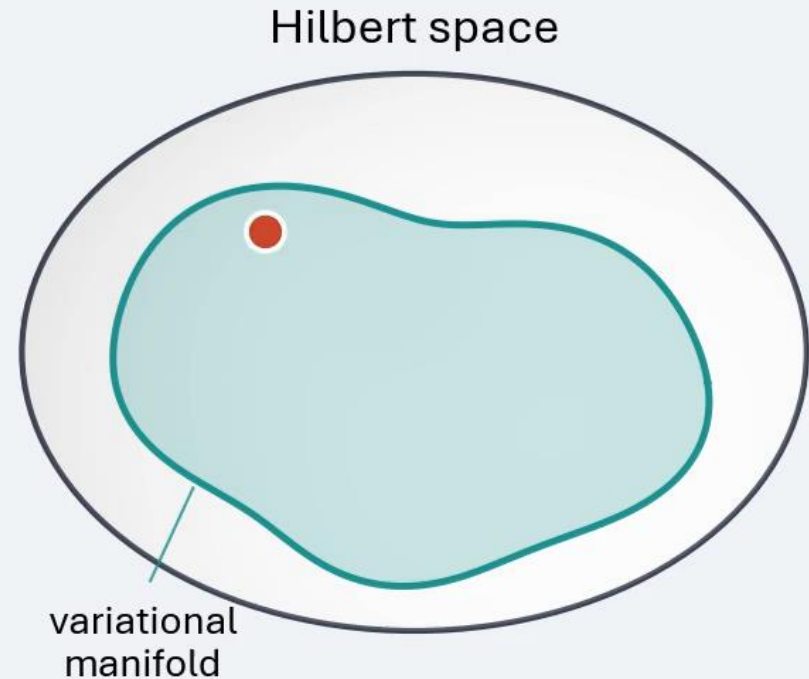
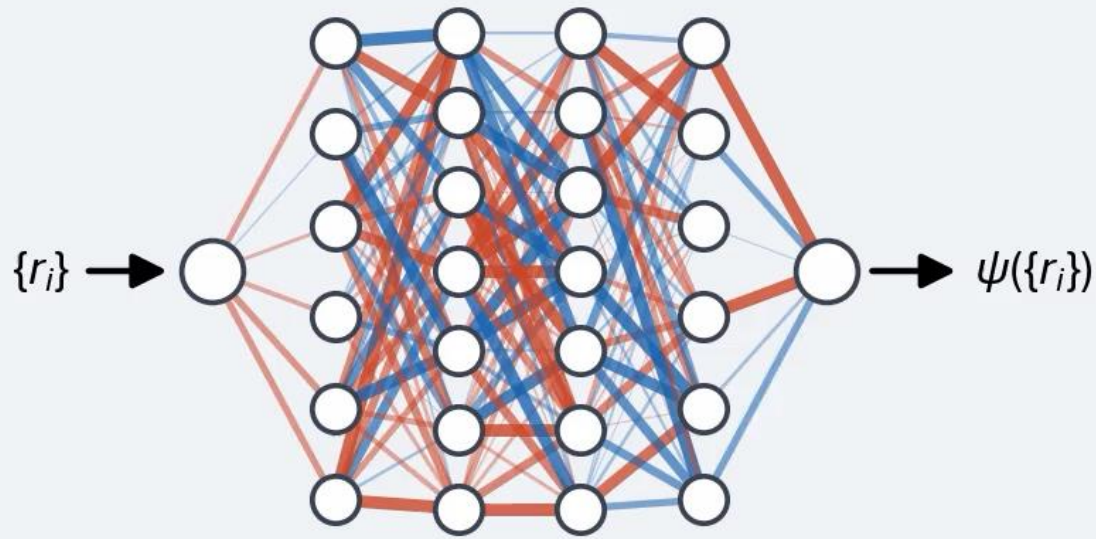
Anyon gas

- Harmonic trap studied to $N = 4$ (Sporre et al, '92)
- Lattice models on a torus, $N = 6$ on 4×5 (Kallin, '93)

Neural quantum states are a significant advancement in our computational capabilities

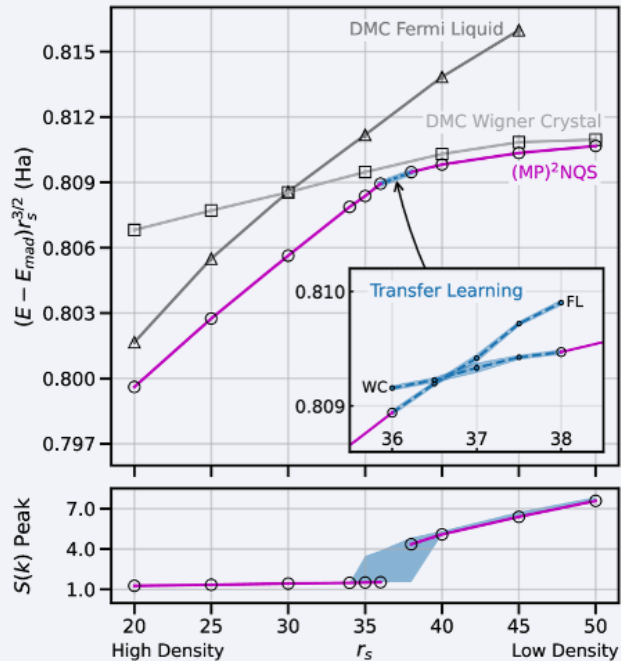
These methods are uniquely suited for studying the anyon gas

Neural quantum states for approximating ground state properties

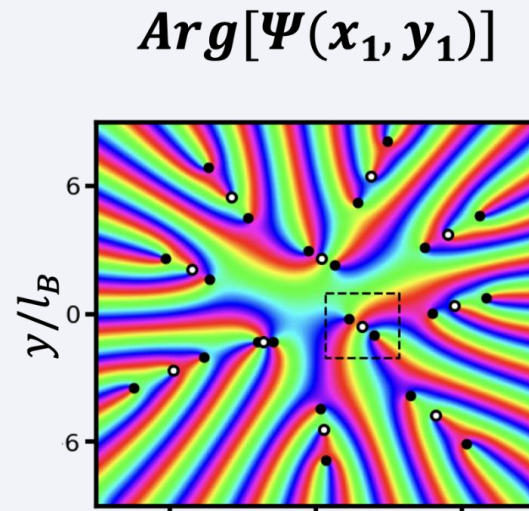


NQS is suitable for anyonic physics

Strongly-correlated continuum systems

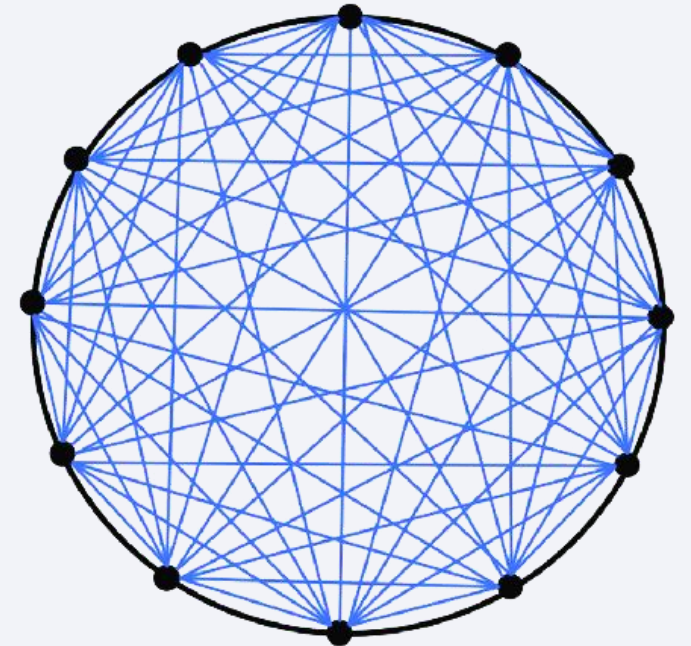


2D electron gas
Smith et al '24



Fractional Quantum Hall
Teng et al '25

Long-range interactions



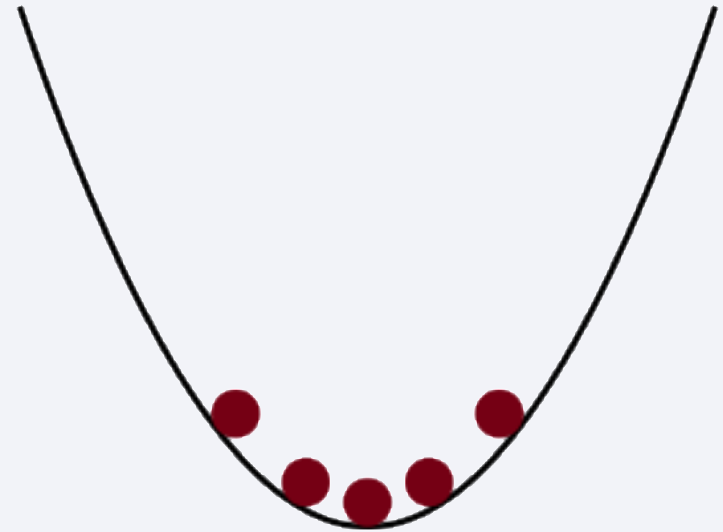
All-to-all disordered Hubbard model
Gauvin-Ndiaye et al '25

Two systems of interest



Lattice anyons on a torus

L. Shackleton, arXiv:2610.xxxxx



Continuum anyons in a harmonic trap

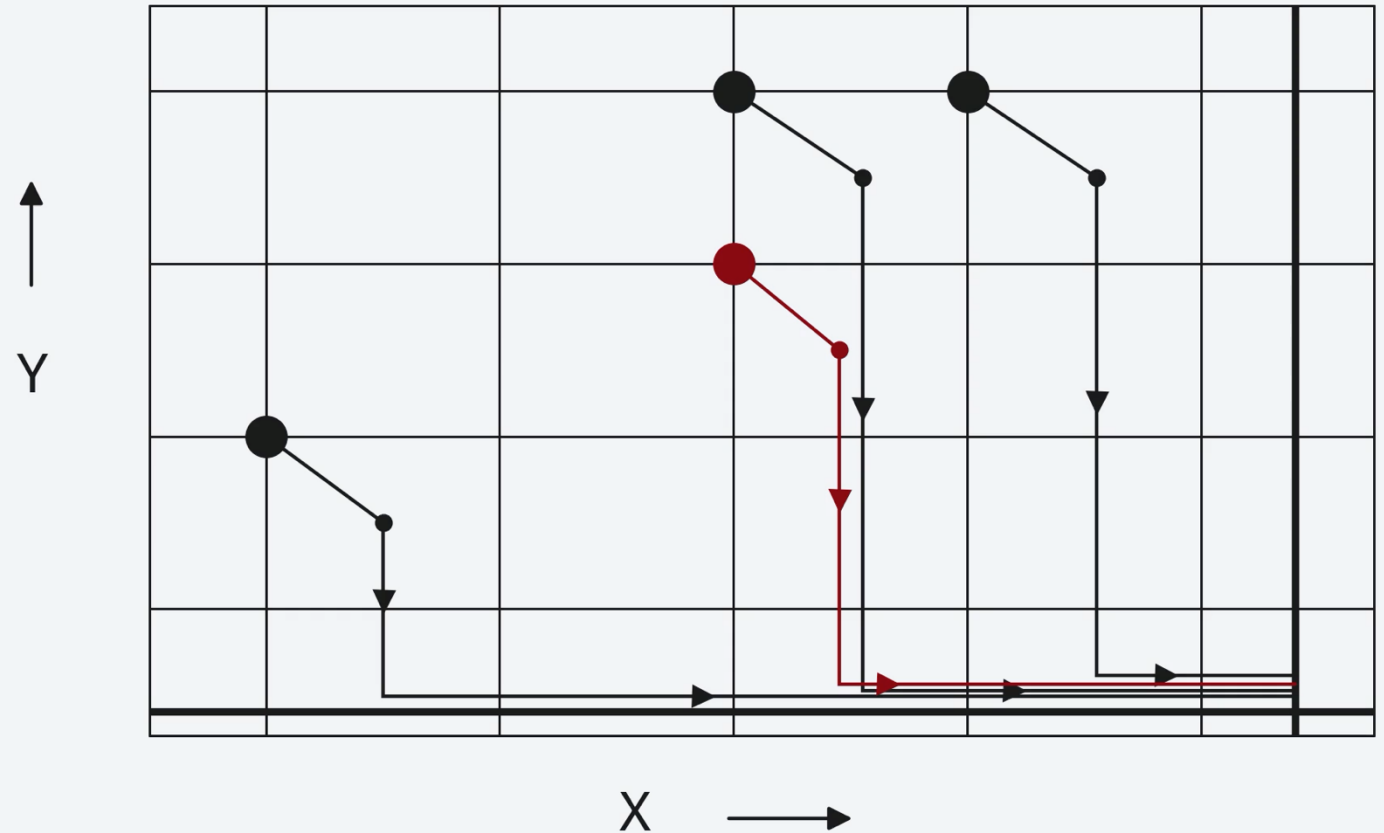
A.P. Fadon, P.M. Tam, L. Shackleton, arXiv:2610.xxxxx

Anyons require multi-component wavefunction

$$|\psi\rangle = |\ell_1, \ell_2, \dots, \ell_N\rangle \otimes |\sigma\rangle$$

Winding around torus
acts non-trivially on σ

Focus on semions ($\alpha = 0.5$), $\sigma = \uparrow, \downarrow$

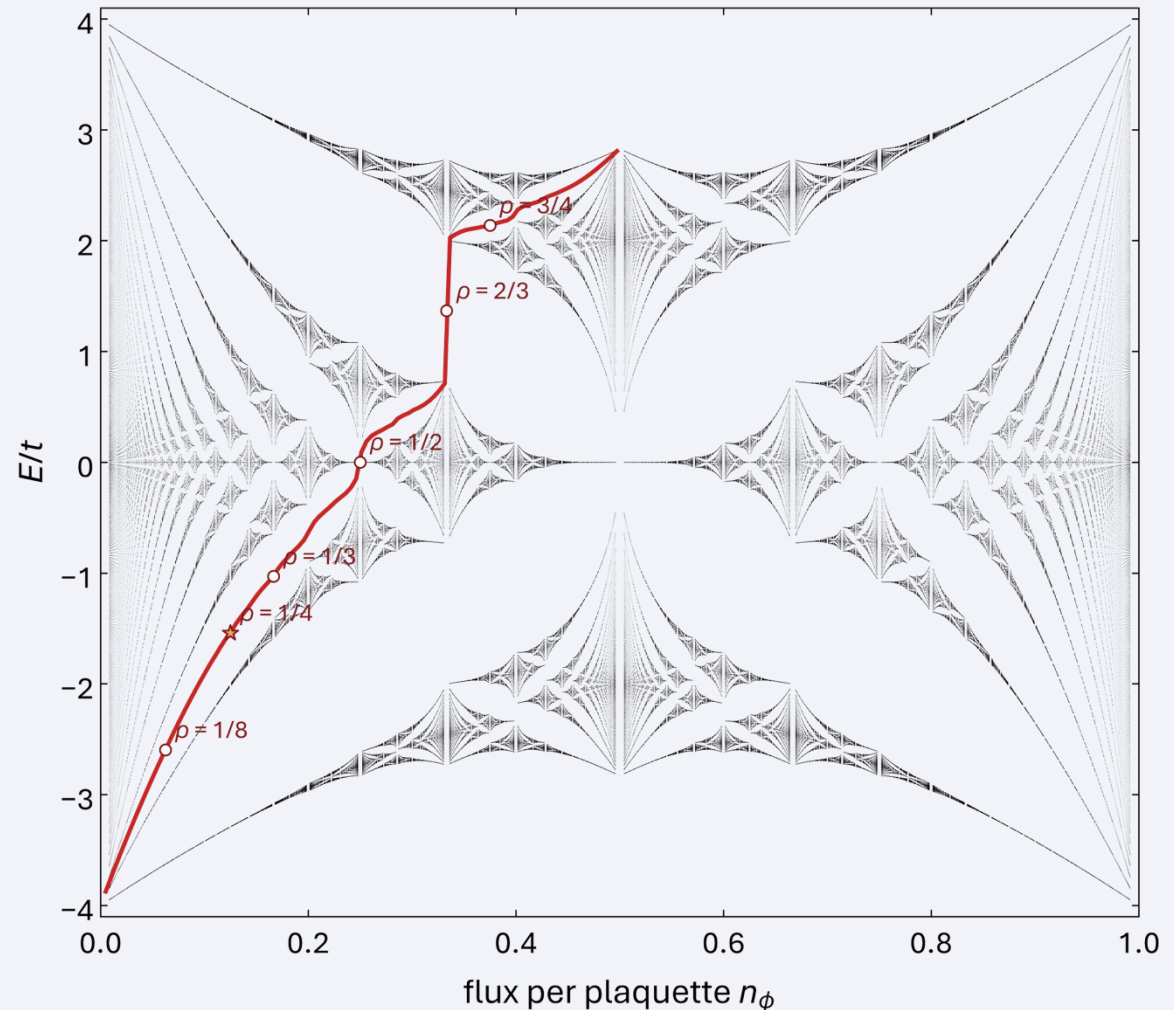


Hamiltonian symmetries, mean-field predictions

$$H = \sum_{\langle ij \rangle} T_{ij} b_i^\dagger b_j$$

Possesses **full space group symmetry**, but complicated action

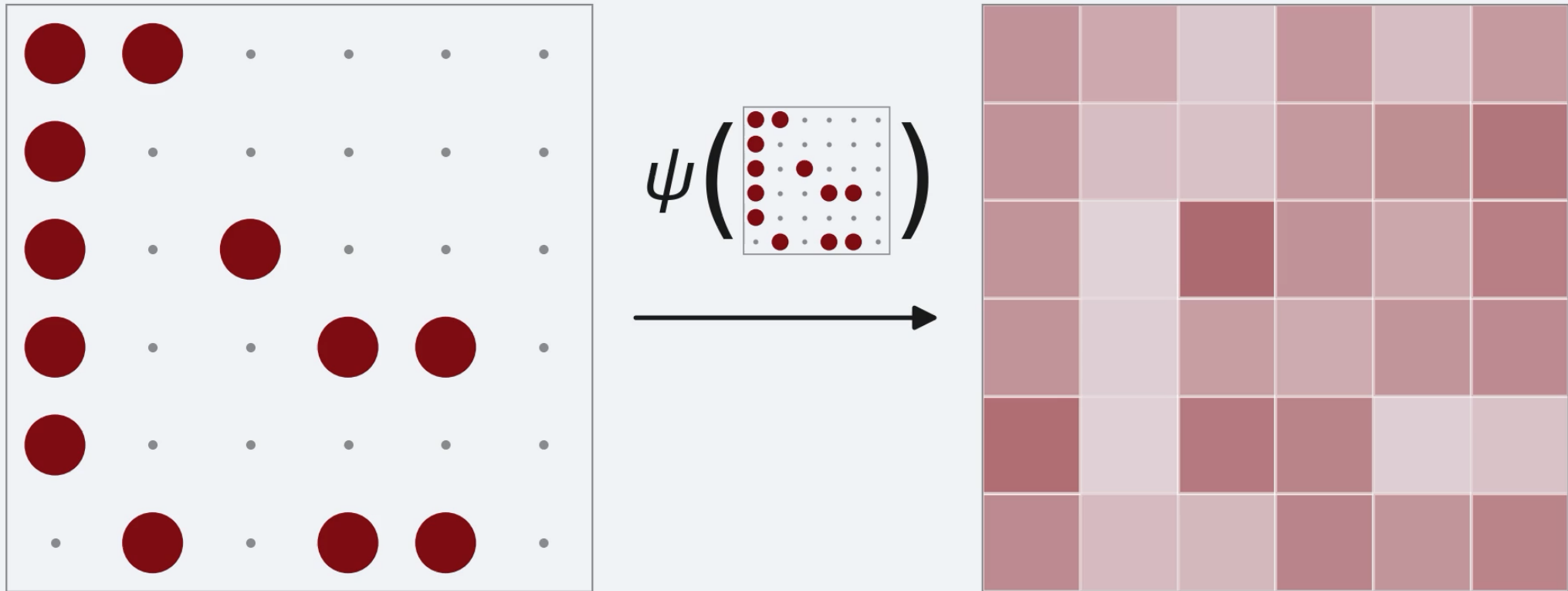
Mean-field predicts superfluid for $\rho < 1/2$, but numerics have only gone to 4 x 5 lattices



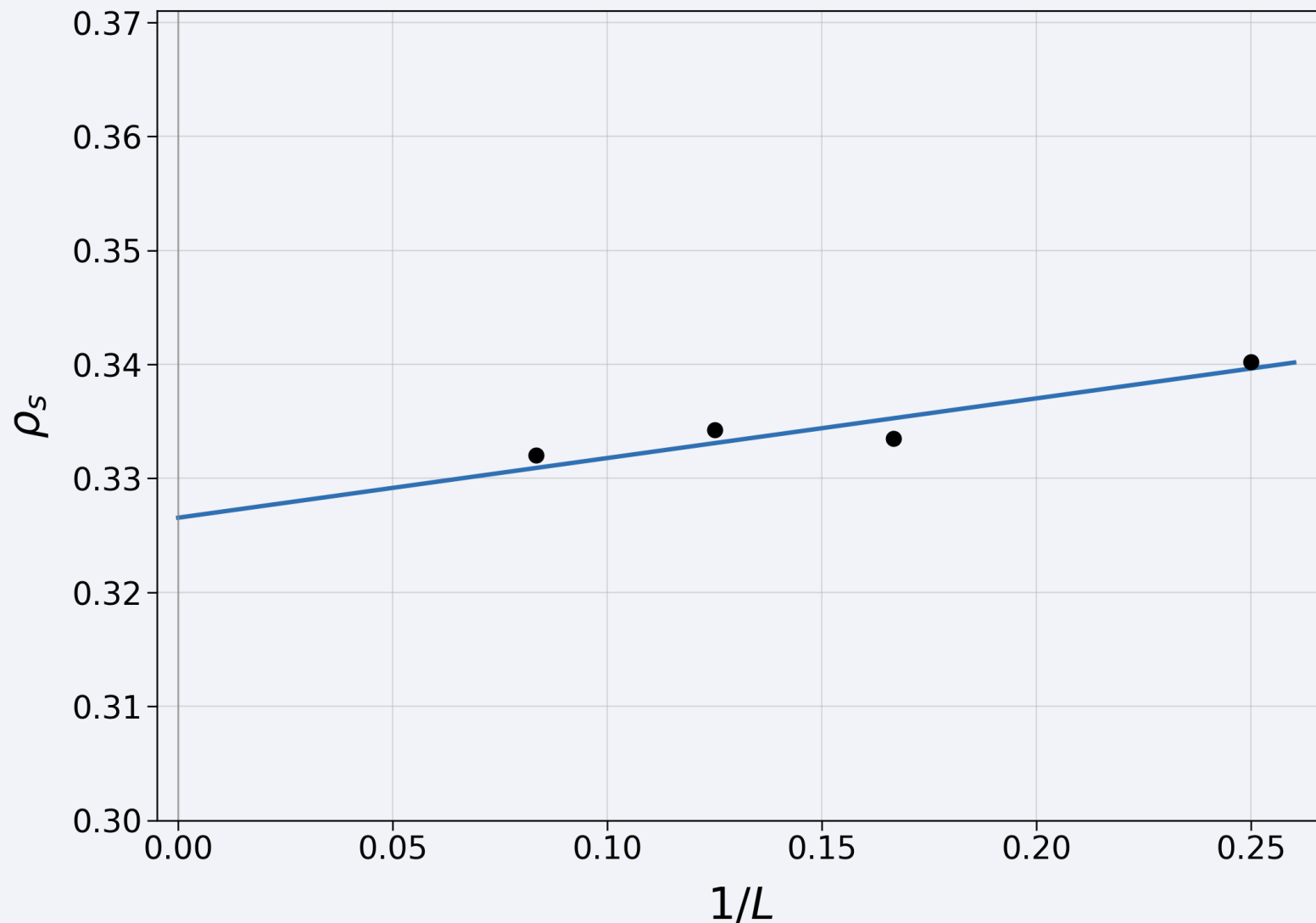
Kallin '93, Feder and Kallin '95)

Group CNNs for enforcing lattice symmetries

$$\tilde{\psi}(x) = \sum_g \chi_g \psi(g^{-1}x)$$



Evidence for superfluidity at $\rho = 1/4$

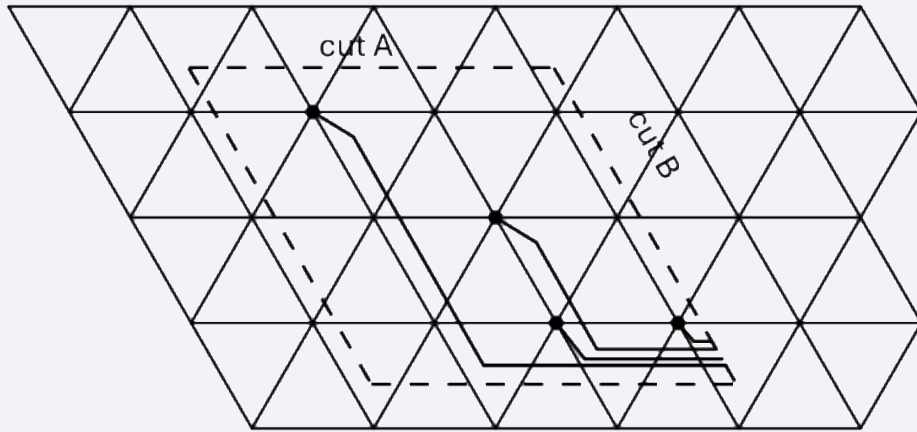


$$\rho_s \sim \frac{\partial^2 E}{\partial \phi^2}$$

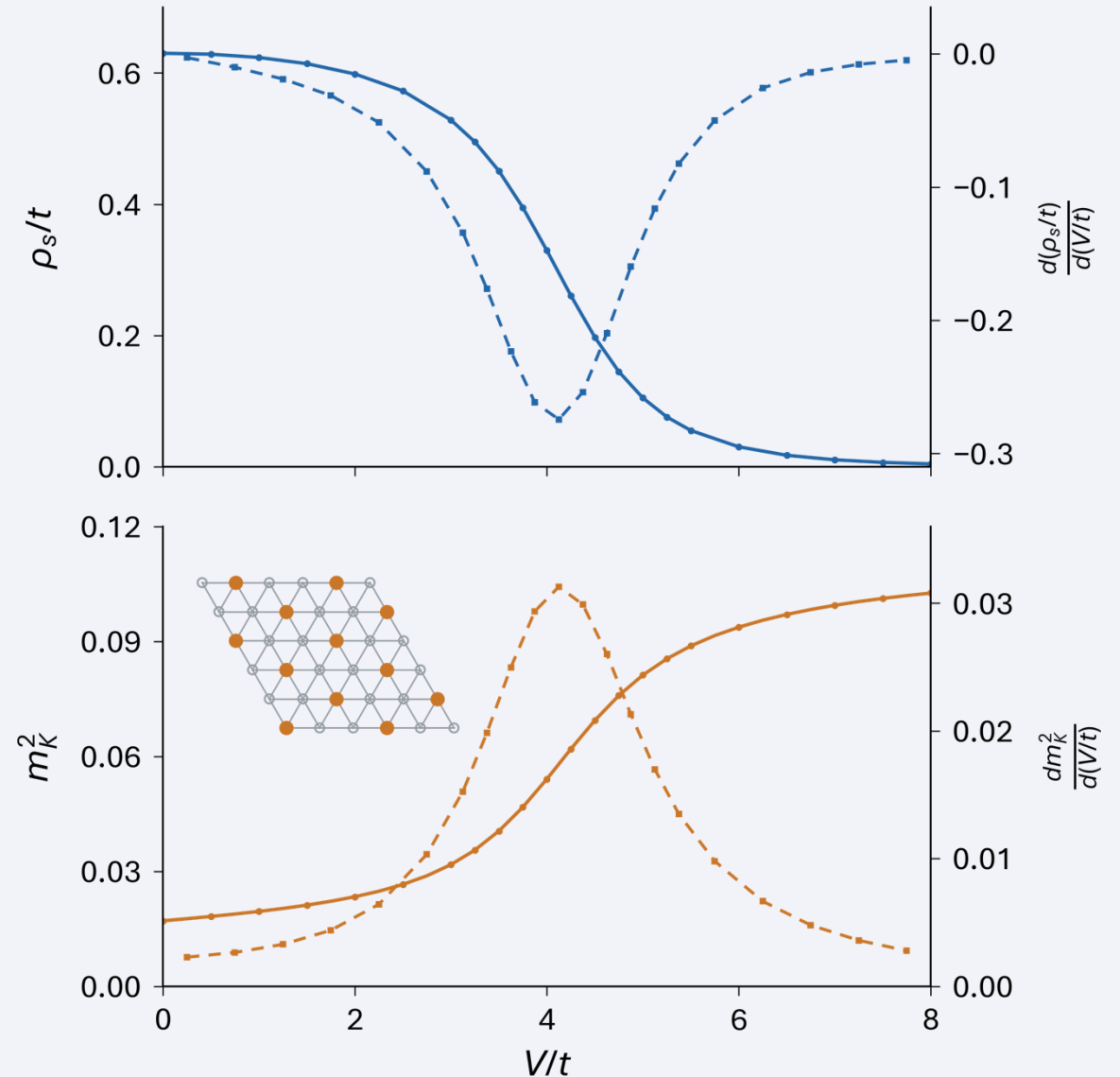
Currently $L = 4, 6, 8, 12$

Triangular lattice, semion crystal at $\rho = 1/3$

$$H = \sum_{\langle ij \rangle} [T_{ij} b_i^\dagger b_j + V n_i n_j]$$



Phase transition at $V/t \approx 4$

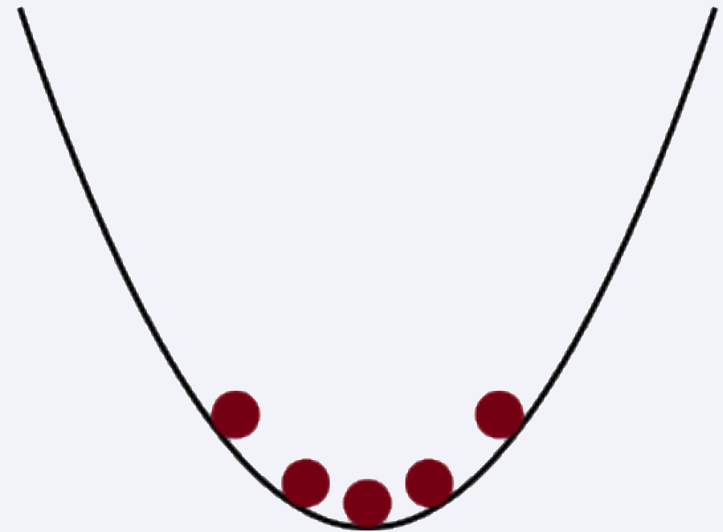


Two systems of interest



Lattice anyons on a torus

L. Shackleton, arXiv:2610.xxxxx



Continuum anyons in a harmonic trap

A.P. Fadon, P.M. Tam, L. Shackleton, arXiv:2610.xxxxx

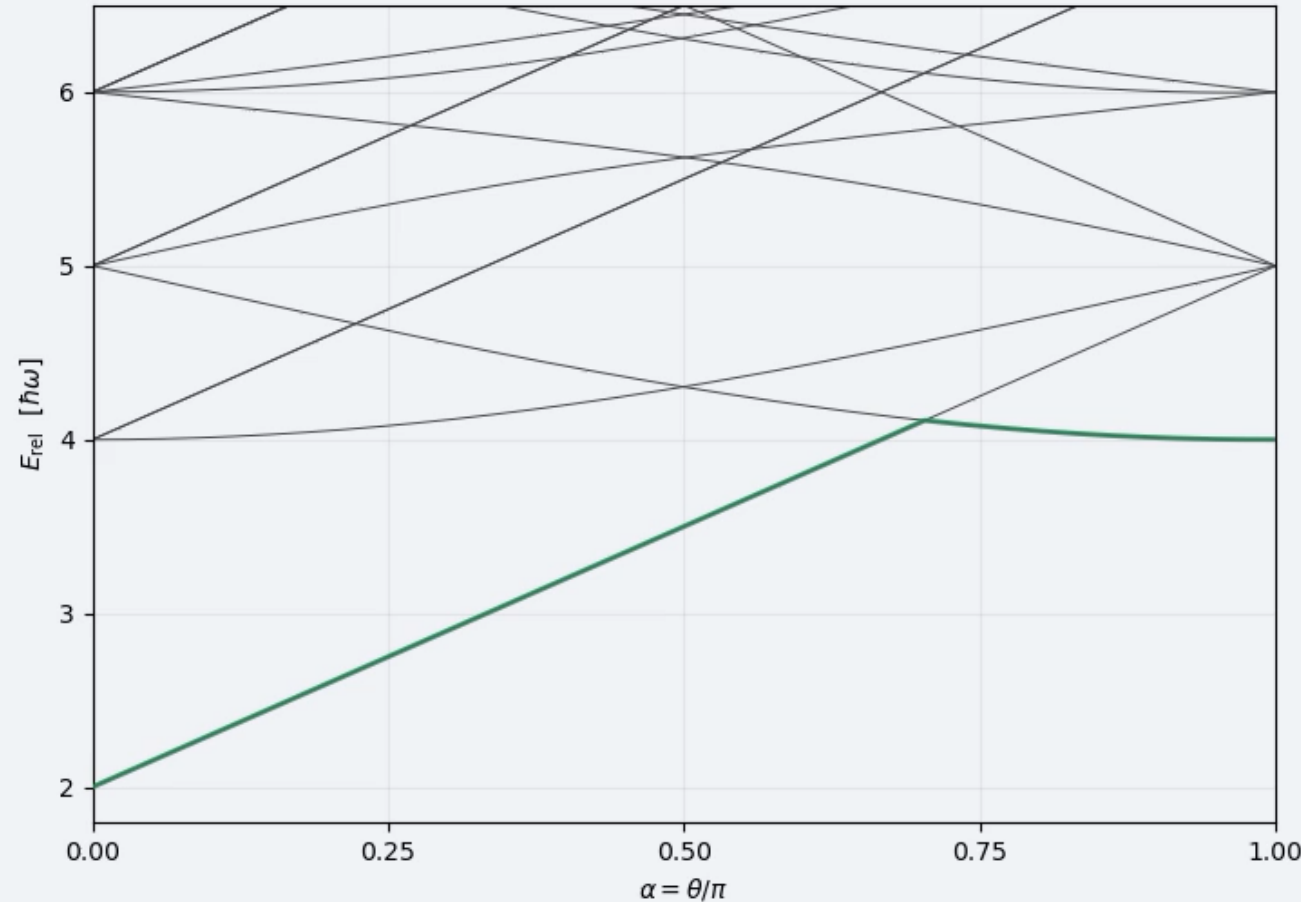
Anyons in a harmonic trap: what is known?

$$H = \frac{1}{2} \sum_{i=1}^N (-i\nabla_i + \mathbf{a}_i)^2 + \frac{\omega^2}{2} \sum_{i=1}^N |\mathbf{r}_i|^2$$

Fix convention: fermionic gauge

Linear branch near $\alpha = 0$

Intrinsically anyonic branch
not accessible to ED



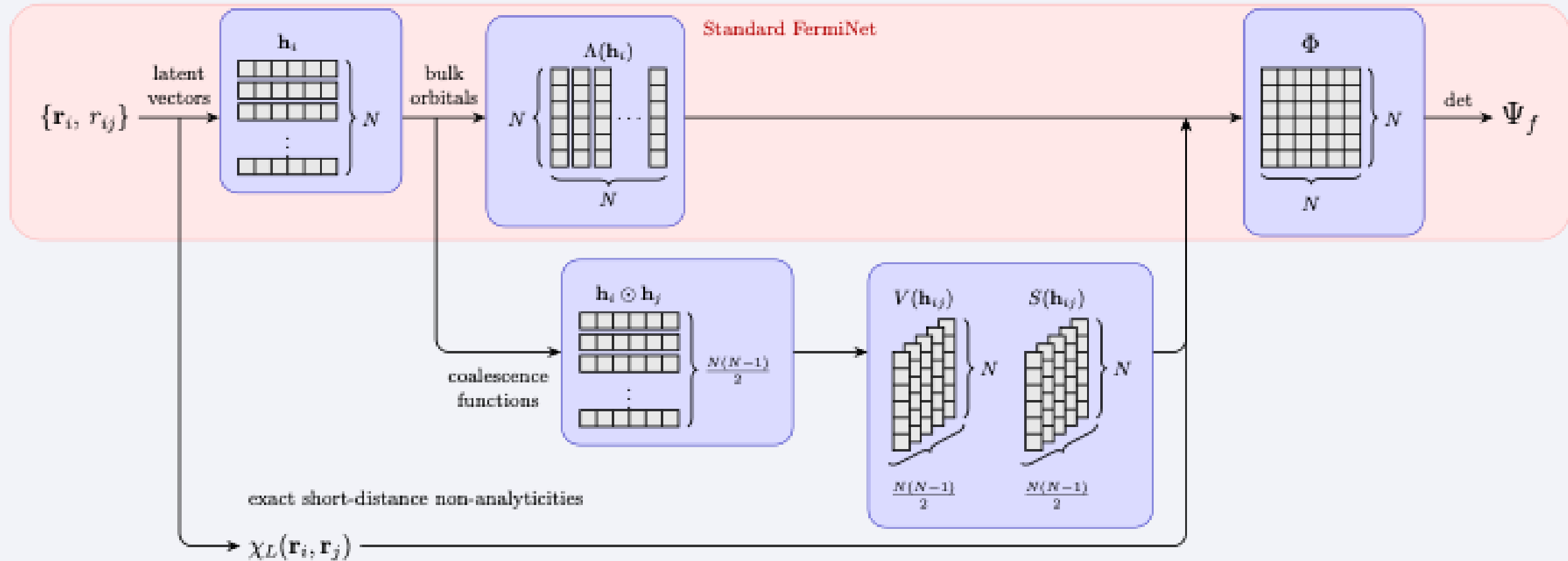
Finite local energy requires multiple short-distance channels

$$\psi(r_{ij} \rightarrow 0) = a z_{ij} r_{ij}^{\alpha-1} + b \bar{z}_{ij} r_{ij}^{1-\alpha} + \dots$$

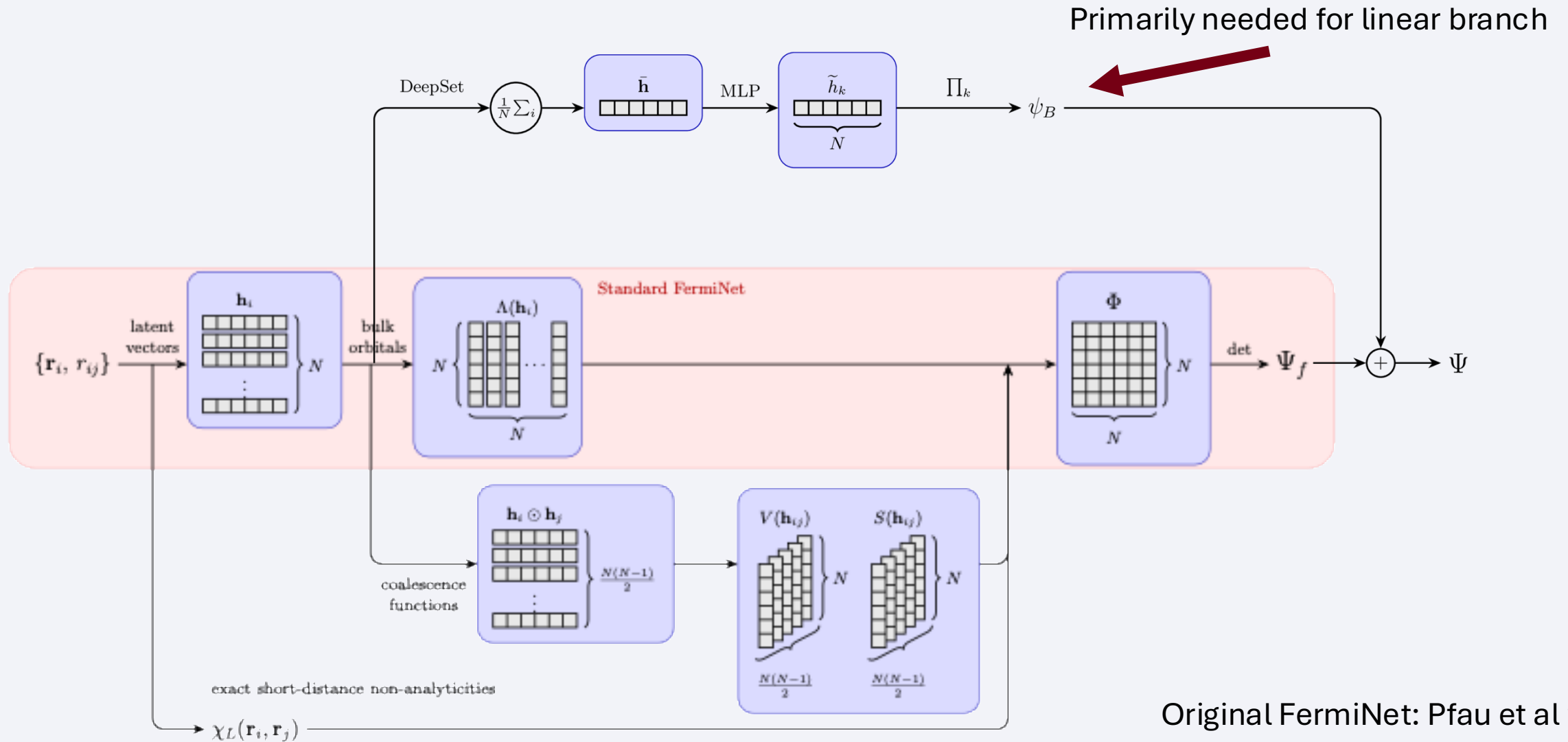
$$z \equiv x + iy$$

- Required for variance to be finite
- Need to get non-analyticities **exactly right** for training to be stable
- Cannot be captured by global Jastrow factor

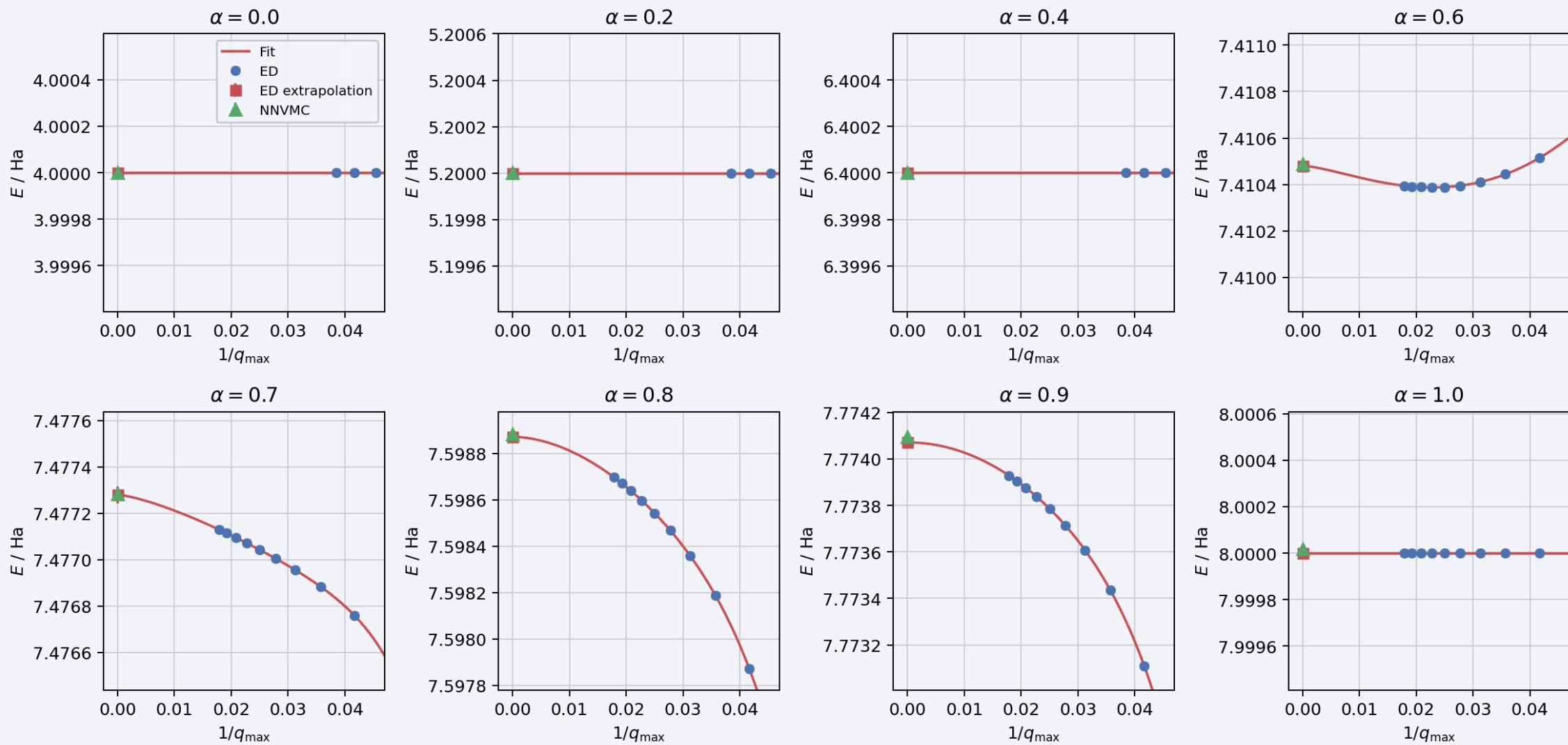
AnyNet: Splicing FermiNet architecture with exact short-distance behavior



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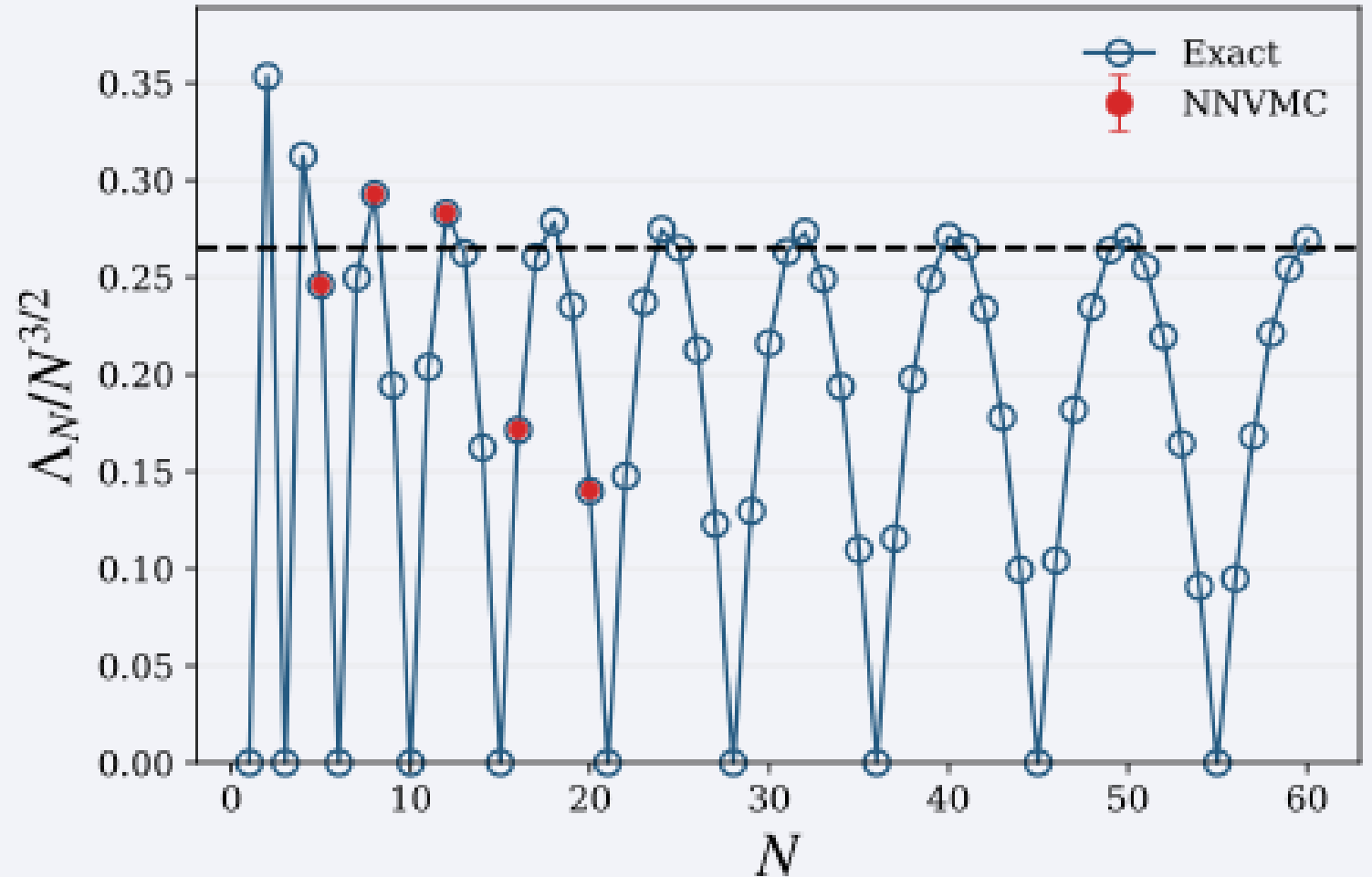


$N = 4$ benchmarks: ED sensitive to cutoff

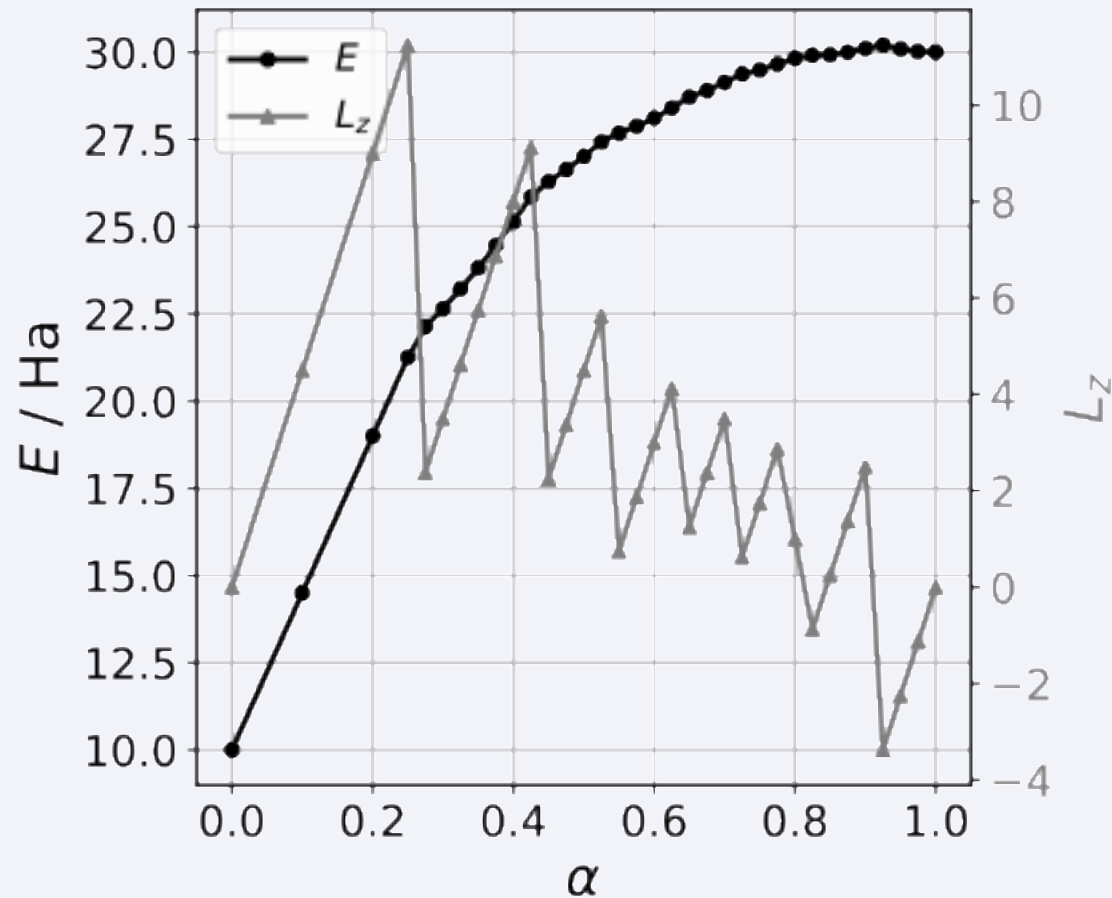


Perturbative agreement in the fermionic limit

$$E = E_F + \Lambda_N(1 - \alpha) + \dots$$

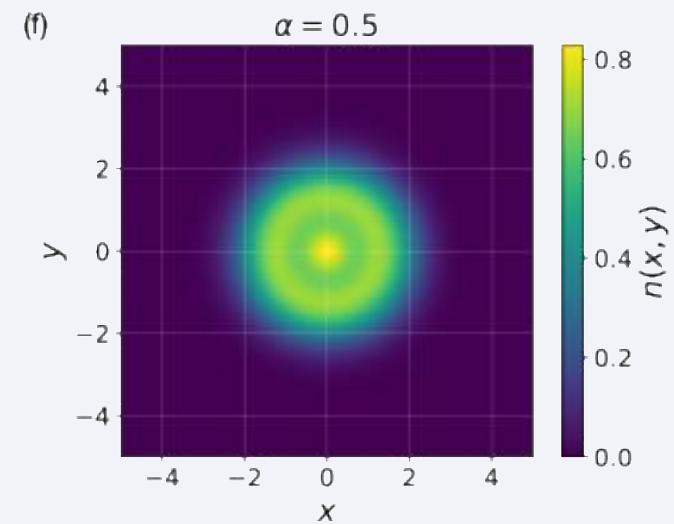
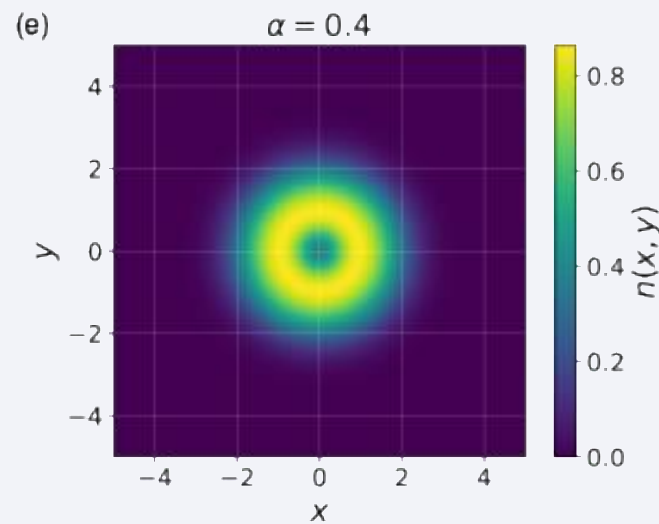
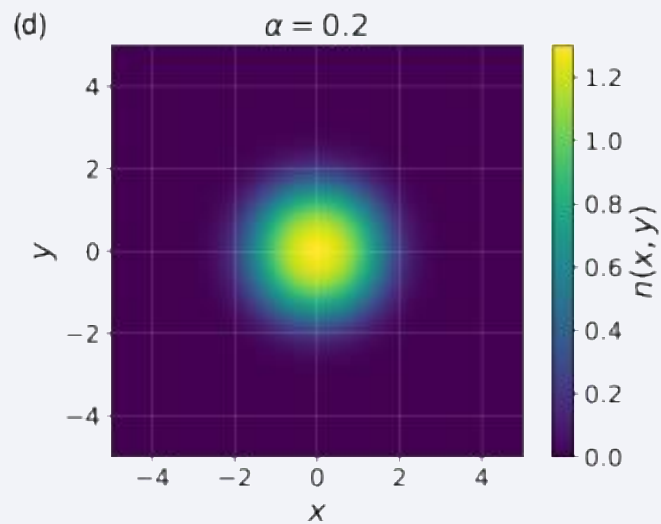
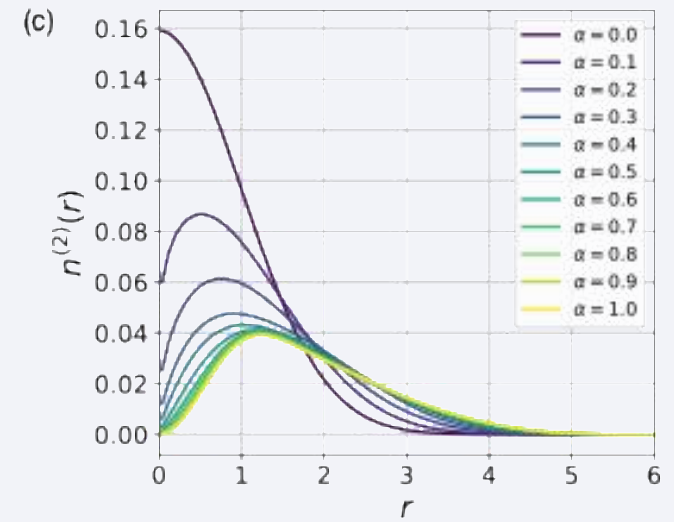
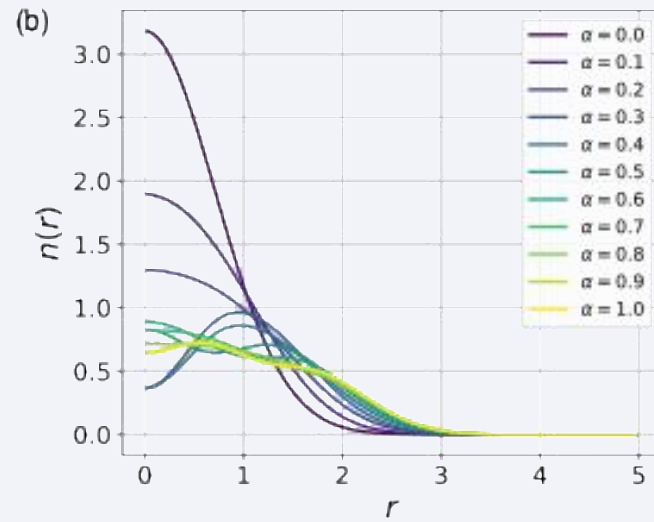
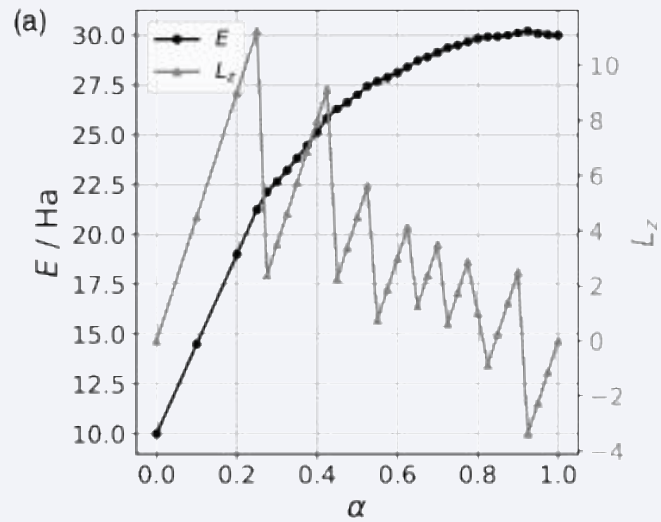


$N = 10$ results



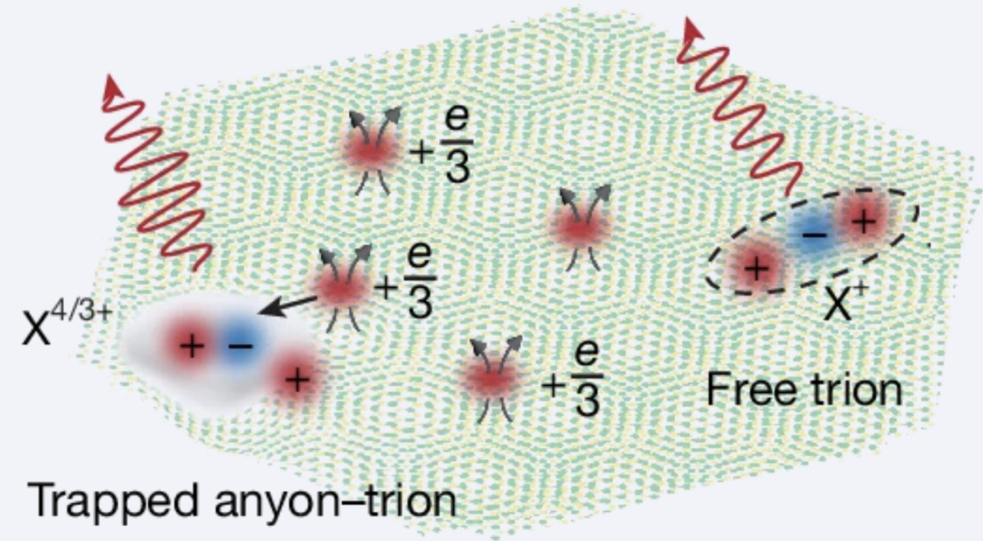
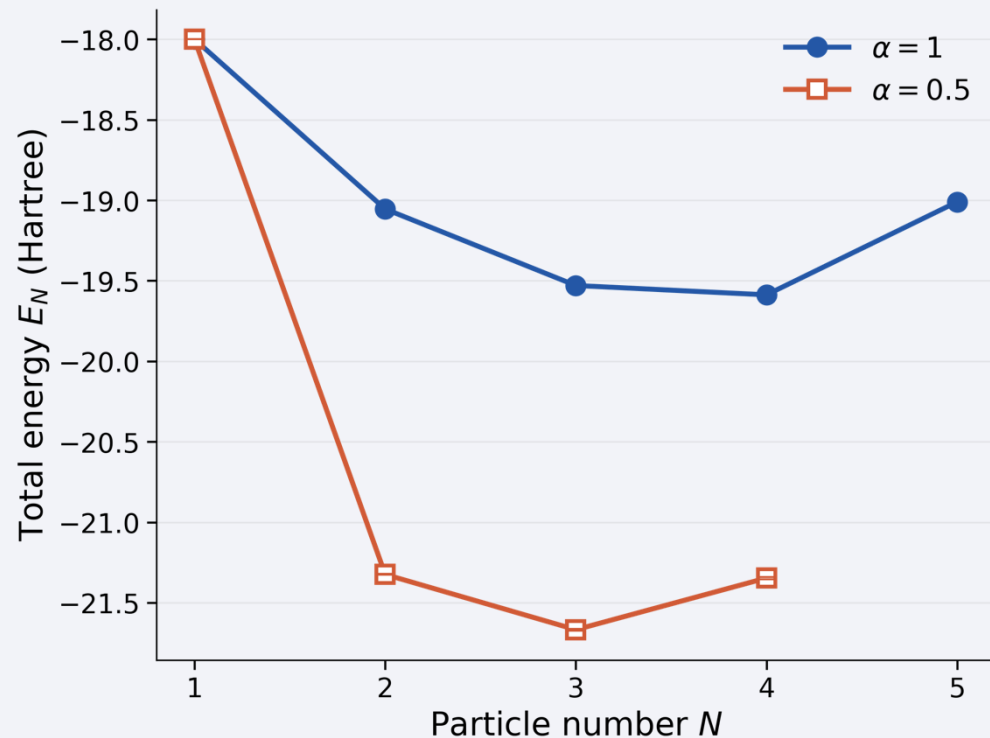
- Clean sequence of level crossings, drops in angular momentum
- $0.25 < \alpha < 0.9$ disconnected from both fermionic and bosonic ends

$N = 10$ results



Additional applications: anyonic quantum chemistry

$$H = -\frac{1}{2m} \sum_i [\nabla_i - i\mathbf{a}_i] - \sum_i \frac{Z}{|\mathbf{r}_i - \mathbf{R}|} + \sum_{i < j} \frac{1}{|\mathbf{r}_i - \mathbf{r}_j|}$$



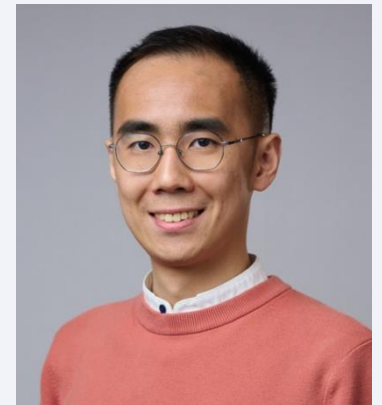
Experiment: Li. *et al.* *Nature* **651**, 48–53 (2026)

Theory: Cui et al, arXiv:2609.30380

Future directions

- Continuum anyons on a torus, sphere, etc
- Larger systems ($N = 16?$)
- Additional probes of superfluidity (ODLRO?)
- Additional interactions, modified dispersions, etc
- Neural wavefunctions for non-Abelian anyons

Neural network wavefunctions put us in a position to resolve the open problem of a free anyon gas



Adapting Slater orbitals to enforce short distance behavior

For each orbital, $\Phi_i = w_{i0} \Phi_{i,bulk} + \sum_{j \neq i} w_{ij} f_{ij}$

Turns on when $r_{ij} \rightarrow 0$



$$f_{ij} = S_{ij} + \Delta_{ij}$$

$$S_{ij} = S_{ji}, \Delta_{ij} = -\Delta_{ji}$$

$$\Delta_{ij} = V_{ij}^+ z_{ij} r_{ij}^{\alpha-1} + V_{ij}^- \bar{z}_{ij} r_{ij}^{1-\alpha}$$

$$\begin{aligned} \det \Phi &\approx \det[S_{12} + \Delta_{12}, S_{21} + \Delta_{21}, \Phi_{3...N}] = \det[S_{12}, \Delta_{12}, \Phi_{3...N}] \\ &= z_{ij} r_{ij}^{\alpha-1} \det[S_{ij}, V_{ij}^+, \Phi_{3...N}] + \bar{z}_{ij} r_{ij}^{1-\alpha} \det[S_{ij}, V_{ij}^-, \Phi_{3...N}] \end{aligned}$$